

Exam Date & Time: 20-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

INTRODUCTION TO MACHINE LEARNING [NPT-4052]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Distinguish between classification and regression. (2)

A)

- B) How is the subset selection method used for dimensionality reduction? (3)

C)

Obtain a linear regression for the data in the following Table assuming that y is the independent variable. (5)

X	3	4	5	6	7
Y	2.5	3.2	3.8	6.5	11.5

- 2) Explain the concept of Linear Discriminant Analysis (LDA) and its purpose in supervised learning. Describe how LDA seeks to reduce dimensionality while maintaining class separability. What is the role of within-class and between-class scatter matrices in LDA? (5)

A)

B)

Consider the following dataset with three classes and two features:

$$\text{Class 1} = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 2 & 3 \end{bmatrix}, \text{Class 2} = \begin{bmatrix} 6 & 5 \\ 7 & 5 \\ 7 & 6 \end{bmatrix}, \text{Class 3} = \begin{bmatrix} 3 & 7 \\ 4 & 7 \\ 5 & 8 \end{bmatrix} \quad (5)$$

Explain how Linear Discriminant Analysis can be used to reduce the dimensionality of this dataset from two features to one, while preserving the maximum class separability.

- 3) Develop a simple perception model and explain learning theory. (5)

A)

- B) Design single layer perceptron with two iterations. Consider the perceptron having with (5)

the initial weights $w_1=0.5$, $w_2 = 0$, learning rate $\alpha=0.2$ and bias $\theta=0.4$ for AND Boolean function. The activation function is the Step function $f(x)$ which gives the output either 0 or 1. If value of $f(x)$ is greater than or equal to 0, it outputs 1 or else it outputs 0.

4) State the key assumptions made by the Naïve Bayes classifier? (2)

A)

B) How is Elbow point used to determine the optimal number of clusters k ? (3)

C)

Cluster the following set of data using k-means algorithm with initial value of objects 2 and 5 with the coordinate values (4, 6) and (12,4) as initial seeds.

Objects	X-coordinate	Y-coordinate
1	2	4
2	4	6
3	6	8
4	10	4
5	12	4

(5)

5) How does Temporal Difference (TD) learning in reinforcement learning differ from Monte Carlo methods? (2)

A)

B) Why is function approximation in reinforcement learning particularly useful for environments with large or continuous state spaces? (3)

C) You are provided with a dataset for a binary classification task that includes the following features and outcomes:

Feature A	Feature B	Outcome
1	3	Yes
2	2	Yes
3	1	No
3	3	No
2	4	Yes

(5)

Explain the process of constructing a decision tree. Describe key concepts such as splitting criteria (Gini impurity, information gain) and stopping conditions.

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